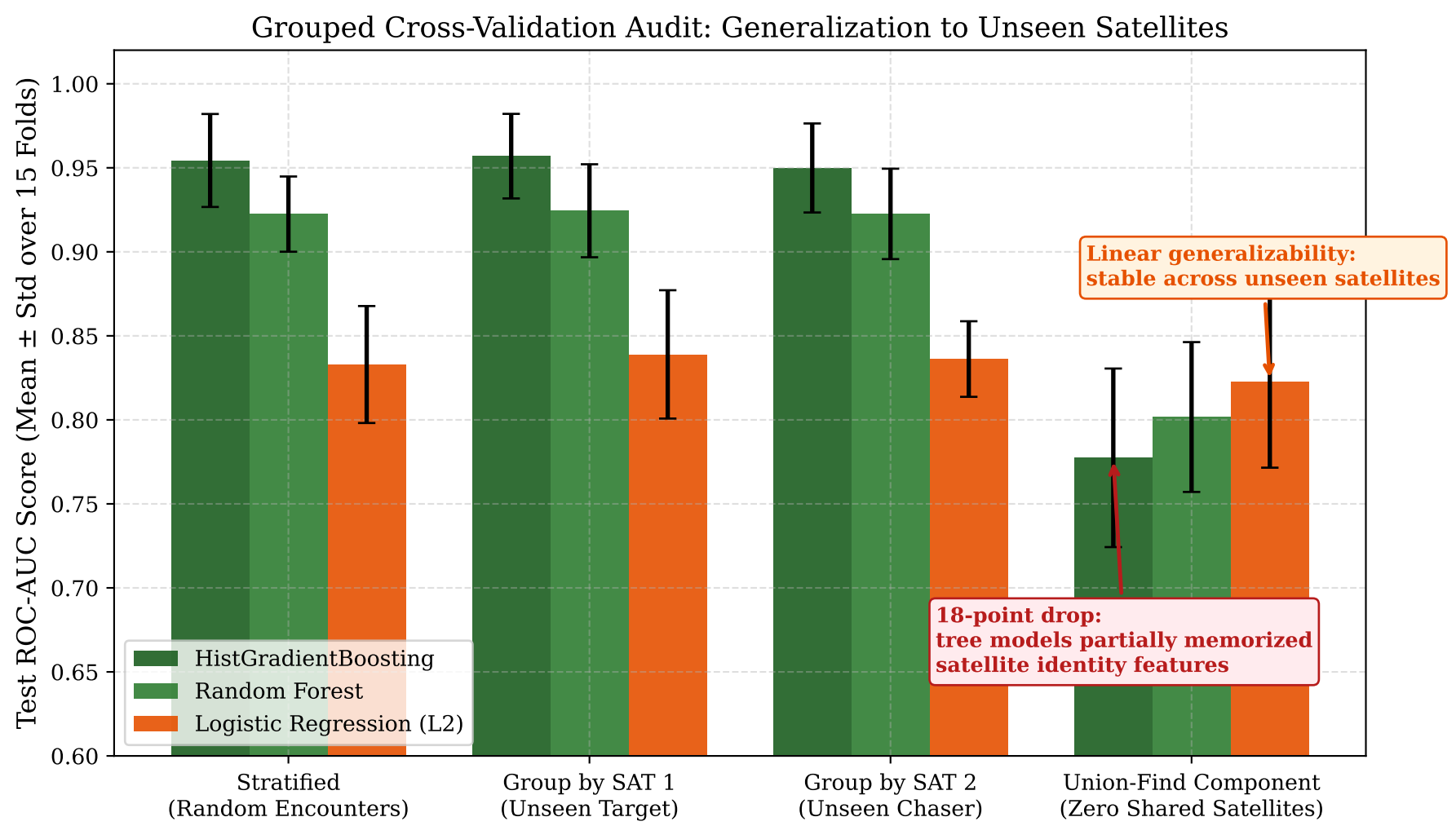
Grouped Cross-Validation Audit: Generalization to Unseen Satellites
Test ROC-AUC Score (Mean ± Std over 15 Folds)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
HistGradientBoosting
Random Forest
Logistic Regression (L2)
Linear generalizability:
stable across unseen satellites
18-point drop:
tree models partially memorized
satellite identity features
Stratified
(Random Encounters)
Group by SAT 1
(Unseen Target)
Group by SAT 2
(Unseen Chaser)
Union-Find Component
(Zero Shared Satellites)